HINAL ARYA

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SUMMARY

AI Engineer Intern at Vena Medical with **2+ years of professional and internship experience** in **machine learning and deep learning** across healthcare, finance, and enterprise settings. Developing **CNNs, ResNet, and LSTM models** for medical imaging and time-series analysis, with **Generative AI, LLM**, and **Retrieval-Augmented Generation (RAG)** exposure using LangChain and FAISS. Skilled in **Python, SQL, PyTorch, and scikit-learn,** with TensorFlow exposure; experienced in preprocessing, feature engineering, model training, evaluation, and deployment using **Docker, REST APIs, and GitHub Actions CI/CD.**

SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, PySpark, ETL, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Statistical Modeling, Hypothesis Testing, A/B Testing, Error Analysis
Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Random Forest, Gradient Boosting, CNNs, LSTMs, Classification, Regression, Clustering, Time-Series Forecasting, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Model Evaluation, Feature Importance Analysis
NLP, Computer Vision & Generative AI: Entity Extraction, Sentiment Analysis, Computer Vision, Medical Imaging, OpenCV, Data Augmentation, LLMs, LangChain, Retrieval-Augmented Generation (RAG), FAISS, Vector Indexing, OpenAI API
Cloud & MLOps: Microsoft Azure, Azure Machine Learning, Azure OpenAI, Azure Blob Storage, Azure Databricks, Docker, MLflow, Git, GitHub Actions, CI/CD, ONNX Runtime, FP16 Conversion, Experiment Tracking, Model Versioning, Model Deployment, Model Monitoring, Data Drift Detection, Automated Testing
APIs & Visualization: Flask, FastAPI, REST APIs, Power BI, Tableau, KPI Dashboards

EXPERIENCE

Vena Medical, Ontario, Canada | *AI Engineer Intern* **January 2026 – August 2026**

- Developed and evaluated **CNNs, ResNet, and LSTM time-series pipelines** using **PyTorch, TensorFlow, and Azure Machine Learning** for intravascular imaging and catheter signal telemetry, improving **feature extraction accuracy by 18%.**
- Built a **medical-document RAG pipeline** using **LangChain, Azure OpenAI, and FAISS vector** indices to query unstructured clinical guidelines, improving **decision-support retrieval precision by 34%.**
- Optimized medical imaging preprocessing and augmentation using **Python, OpenCV, and NumPy**, reducing **execution time by 42%** while supporting software validation compliance; contributed to Docker, **PyTorch model** checkpoint, and **GitHub Actions CI/CD** workflows for automated testing and model versioning, shortening **deployment cycles by 35%.**

Fortune Infosys, India | *Data Scientist* **November 2021 – December 2023**

- Developed **classification, regression, and clustering models using Scikit-learn, XGBoost, and Random Forest** on **500K+ financial records**; deployed supervised models through **Azure Machine Learning**, applied hyperparameter tuning, cross-validation, model evaluation, and feature importance analysis, and communicated insights to stakeholders, **improving evaluation metric by 22%.**
- Used **Python, SQL, Pandas, NumPy, and Azure Blob Storage** for data access and preparation; performed data cleaning, preprocessing, **EDA, feature engineering, statistical modeling, hypothesis testing, and A/B testing** to address business requirements.
- Used **MLflow for experiment tracking and model versioning**; performed error analysis and monitored **model performance and data drift** to guide retraining and maintain prediction reliability.

Fortune Infosys, India | *Data Scientist Intern* **June 2021 – November 2021**

- Implemented and evaluated **Random Forest and Gradient Boosting models** using Python and SQL for **customer churn prediction**, achieving **0.88 ROC-AUC** and supporting retention initiatives associated with a **15% reduction in quarterly customer attrition.**
- Contributed to **ETL and feature engineering** improvements using **Pandas, NumPy**, **PySpark in Azure Databricks**, reducing **weekly data preparation time by 28%**; built **Power BI, Tableau dashboards** to track model KPIs, improving **operational reporting speed by 40**%.

EDUCATION

Master of Applied Computing **April 2025**
Wilfrid Laurier University, Waterloo, Ontario, Canada
Bachelor of Engineering, Information and Communication Technology **June 2021**
Gujarat Technology University, Gujarat, India

PROJECTS

AI-Powered IC50 Drug Response Forecasting **January 2025 – April 2025**
Tech Stack: Python, scikit-learn, XGBoost, TensorFlow, Flask, Docker

- End-to-end system predicting **cancer cell drug sensitivity** from genomic and compound features; Benchmarked **Random Forest, XGBoost**, **and a custom neural network** across engineered, cross-validated feature pipelines, evaluated via **F1-score, ROC-AUC, and RMSE**; Deployed as a **Flask REST API in Docker** for real-time drug-response predictions from submitted compound data.

Intelligent Climate Trend Prediction Using LSTM **September 2024 – December 2024**
Tech Stack: Python, TensorFlow, Keras, Pandas, NumPy

- **LSTM-based forecasting system** for temperature, precipitation, and humidity from multi-decade climate data, with sequence-length, dropout, and learning-rate tuning for stability. Evaluated with **RMSE/MAE** and validated generalization across seasons and regions.